

ADRENAL INSUFFICIENCY

AI

CAUSES

TEST

HPA axis

TREATMENT

Tertiary

Mass Lesions
XRT
Infiltrative
Infectious
Infarction
Trauma/Surgery

Positive Cort Stim
↓ ACTH
(even after CRH test)

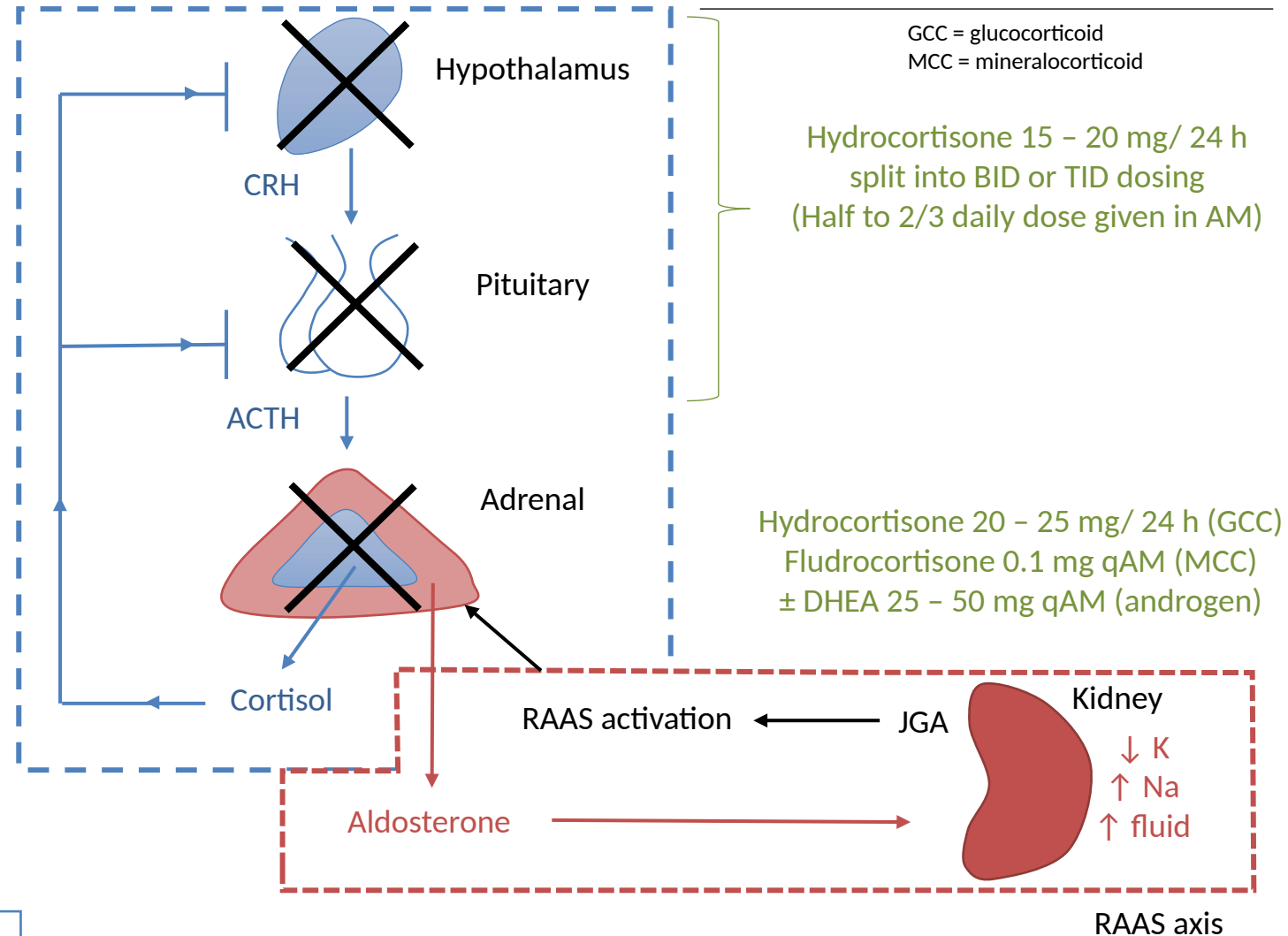
Secondary

Positive Cort Stim
↓ ACTH

Primary

Autoimmune adrenalitis
Infectious adrenalitis
Tumor/ Metastases
Hemorrhage
Drugs

Negative Cort Stim
↑ ACTH
↓ aldosterone
↑ renin



Click on the “AI buttons” to advance through the causes, tests and treatments for each.

8am Cortisol to diagnose:
<3 – ruled in
<10 – suspected
>18 – ruled out

Cortisol stim test:
1. ~ 8a cortisol
2. ~ 8a cosyntropin dose
3. Cortisol level @ 60min

Advance slides to review hormone pathways for HPA Axis and RAAS Axis

Practice Update — Cortisol Assays & GC-Induced AI

- **Cortisol cutoffs are ASSAY-SPECIFIC.** The classic 8 AM cortisol $>18 \mu\text{g/dL}$ "rules out" and the ACTH-stim peak $\geq 18 \mu\text{g/dL}$ come from older polyclonal assays.
- Modern specific monoclonal immunoassays and LC-MS/MS read ~20–30% lower → use a peak/rule-out threshold of ~14–15 $\mu\text{g/dL}$, not 18. Always apply your own lab's assay-specific cutoff.
- Diagnosis algorithm unchanged: 8 AM cortisol first; if low, ACTH distinguishes primary (high ACTH) from central (low/normal ACTH). Standard test = 250 μg cosyntropin, cortisol at 30 or 60 min.
- Most common cause of AI overall = exogenous glucocorticoids (tertiary). The 2024 ESE/Endocrine Society joint guideline gives a structured approach to diagnosing and tapering glucocorticoid-induced AI.
- Adrenal crisis = treat empirically: hydrocortisone 100 mg IV bolus, then 200 mg/24h (or 50 mg IV q6h) + IV fluids; do not wait for confirmatory testing.

Sources: Endocrine Society primary-AI guideline (JCEM 2016); assay-specific cutoffs (J Endocr Soc 2021, bvab022); ESE/ES glucocorticoid-induced AI guideline (JCEM 2024).

Take-Home Points

- PRIMARY AI (adrenal) causes adrenal crisis more readily than central (secondary/tertiary) because mineralocorticoid (aldosterone) is also lost → hypovolemia, hyperkalemia, hyponatremia. Central AI spares the RAAS axis.
- Diagnose with an 8 AM cortisol: very low rules in, clearly high rules out; indeterminate values need an ACTH (cosyntropin) stimulation test. Use your lab's assay-specific cutoff (~14–15 µg/dL with modern assays).
- Adrenal crisis = hypotension/shock ± fever, abdominal pain, nausea, fatigue. It is an emergency — give stress-dose hydrocortisone and fluids immediately, before diagnostic confirmation.
- Replacement: hydrocortisone (split dosing, larger AM dose); add fludrocortisone for primary AI only (mineralocorticoid); ± DHEA. Uptitrate ("stress dosing") during illness/surgery.

Attribution & Optimization Notes

- Adapted from TeachIM.org "Adrenal Insufficiency" (Yilin Zhang, MD; Brandon Fainstad, MD; expert review pending). Published Oct 2020.
- teachim.org/teaching_material/adrenal-insufficiency/ — Licensed CC BY-NC 4.0 (attribution, non-commercial). Educational use only; clinician-reviewed.
- No factual deck errors found — HPA/RAAS physiology and crisis teaching verified current. Append-only optimization (no edits to the original animated board).
- Added (source-cited): assay-specific cortisol cutoffs (modern assays ~14–15 vs older 18 µg/dL), 2024 ESE/ES glucocorticoid-induced AI guideline, explicit adrenal-crisis stress-dosing.
- Note: the board lists 8 AM cortisol ">18 ruled out"; the TeachIM lesson page take-home says ">15". Both are assay-dependent — apply local lab cutoffs. Companion files: Learner board PPTX, handout PDF.
- Optimization run 2026-06-10. Medical content should be clinician-reviewed before teaching.